Final Results: Super-Emitter Benchmark (4–8 masts)

CH4-T testbed, measured wind, 90% conformal coverage

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Benchmark. Sources $q \sim \mathrm{LogUniform}(100, 500)\,\mathrm{kg\,h^{-1}}$ (all super-emitters), $S \in \{4, \ldots, 8\}$ masts, 10,000/1,000/2,000/2,000 train/val/calib/test scenarios, anemometer error 10°/10% per step; no model observes the true wind. Method: three physics input images (average evidence, wind spread of evidence, sensor visibility; asinh-stabilized, one shared set of K=8 error-model draws), zero-initialized convolutional head joining the base network at half the training budget. Split conformal at 90%. Exact-model reference under known wind: 4.4 m.

Table I: localization performance

Ranges are min–max over three training seeds. The physics-only row uses the identical wrapper, calibration scenarios, rate prior, and K=8 wind budget.

Method	Coverage	Radius (m)	Below 50 m
Wind-averaged physics only	1.000	282 (full site)	0%
DeepSets, no maps	0.887–0.914	97–100	0.0–0.1%
DeepSets + maps	0.890–0.916	**73–75**	**16–17%**
GNN, no maps	0.896–0.904	68–69	1–7%
GNN + maps	0.889–0.911	**61–62**	**27–29%**
Set Transformer, no maps	0.891–0.901	72–75	8–12%
Set Transformer + maps	0.887–0.906	**64–65**	**28–30%**

Paired effects (seed 1, same 2,000 scenarios, 10,000 paired bootstrap resamples, 95% CIs):

Base network	ΔRadius (m)	Δ(<50 m) (pp)	Smaller region
DeepSets	−24.6 [−25.8, −23.0]	+16.2 [+14.6, +17.9]	95.9%
GNN	−5.2 [−6.4, −4.2]	+20.5 [+18.6, +22.3]	83.2%
Set Transformer	−10.7 [−12.1, −9.5]	+20.9 [+19.1, +22.7]	87.4%

Every interval excludes zero on both metrics for all three architectures.

Table II: incremental effect of the physics images (DeepSets)

Three seeds per row. The evidence and visibility rungs are computed at the single measured wind; the final rung introduces the K=8 error-model draws, adding the wind-spread image and draw-averaging the first two.

Model configuration	Radius (m)	Below 50 m
Input-only network	97–100	0.0–0.1%
+ Source-evidence map	75–78	11–12%
+ Sensor-visibility map	74–76	12–14%
+ Wind-spread map (K=8)	**73–75**	**16–17%**

Both metrics improve monotonically; the source-evidence image contributes the largest single step (≈ -22 m).

Provenance

All values recomputed from per-scenario audit arrays (`pw_audit_*.npz`, sizes + covered + masks) in `csr-data-se48`; paired CIs in `results/se48_table_deltas_ci.json`. Branch `super-emitter`; every prior configuration (6–12 and 2–8 populations, clipped evidence map, joint head training) is archived with its results. Figures: `fig_data_dist` (region-radius distributions), `fig_data_masts` (median radius vs. mast count), both seed 1.